Why Average and Difference Work

The Mechanism Behind the Symmetric–Antisymmetric Split, from Two Numbers to Clifford's Product

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What this deck is for

Tuesday's deck *used* the decomposition

$$A = \tfrac{1}{2}(A + A') + \tfrac{1}{2}(A - A')$$

on functions, matrices, and products of vectors. The history deck traced where it came from. This deck answers a different question:

Why does this work?

What is it about "take the average, take half the difference" that makes a two-unknown problem fall apart into two one-unknown problems, makes cross terms cancel, and – in Hamilton's and Clifford's hands – makes a single product of two vectors separate into a scalar and something else?

The answer is elementary, and it is the same at every level:

1. **Average and difference are coordinates** – a 45° rotation of the (x, y) plane.
2. **The swap is a reflection**, and these coordinates are its *eigencoordinates*: one is fixed, the other is negated.
3. **Anything symmetric in the two things is even in the difference coordinate**, so odd powers cancel. That is where the simplification comes from.
4. **For products**, the "two things" are ab and ba; the same three facts give $ab = a \cdot b + a \wedge b$.

Roadmap

1. Two Numbers, Two Coordinate Systems
2. From Two Numbers to Anything with a Swap
3. Hamilton: The Pair (**uv**, **vu**)
4. Clifford: The Pair (ab, ba)
5. Summary

Two Numbers, Two Coordinate Systems

A pair of numbers is a point in the plane

Take two numbers, $x = 7$ and $y = 3$. Think of them as the point $(x, y) = (7, 3)$.

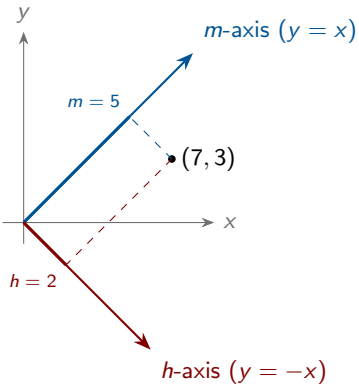
Now compute

$$m = \frac{x + y}{2} = 5, \qquad h = \frac{x - y}{2} = 2.$$

These are *also* two numbers, and together they determine the original pair:

$$x = m + h = 7, \qquad y = m - h = 3.$$

So (m, h) is a second name for the same point. Nothing is lost, nothing is added.



The picture

m is how far the point lies along the diagonal $y = x$; h is how far it lies off that diagonal, toward the x side. The (m, h) axes are the (x, y) axes rotated by 45° (and rescaled by $\sqrt{2}$).

The change of coordinates as a matrix

$$\begin{pmatrix} m \\ h \end{pmatrix} = \underbrace{\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}}_{T} \begin{pmatrix} x \\ y \end{pmatrix}, \qquad \begin{pmatrix} x \\ y \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}}_{T^{-1}} \begin{pmatrix} m \\ h \end{pmatrix}.$$

- $T^{-1} = 2T$: the transformation is its own inverse up to a factor of 2. Going from (x, y) to (m, h) and going back are the *same* operation.
- Rescale by $\sqrt{2}$ to make it exactly self-inverse and orthogonal:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad H^2 = I, \qquad H^{\mathsf{T}} H = I.$$

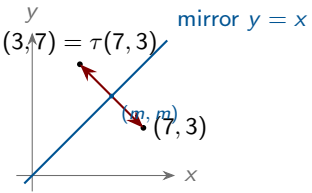
  H is a reflection of the plane; it sends the x-axis to the diagonal and the y-axis to the anti-diagonal.
- The rows of T – $(\frac{1}{2}, \frac{1}{2})$ and $(\frac{1}{2}, -\frac{1}{2})$ – are **orthogonal**. So are the columns of T^{-1}. That is why the two coordinates never interfere with each other.
- **Why the factor $\frac{1}{2}$:** it makes m the *mean* and h the *deviation from the mean*, with $x = m \pm h$ free of stray constants. With $\frac{1}{\sqrt{2}}$ instead, lengths are preserved. Both conventions appear; the structure is identical.

The swap is a reflection

The operation that exchanges the two numbers, $\tau\colon (x, y) \mapsto (y, x)$, $\tau = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, is a **reflection** of the plane across the diagonal $y = x$; $\tau^2 = I$.

What does τ do to m and h?

$$m' = \tfrac{1}{2}(y + x) = m \qquad \text{(unchanged)}$$
$$h' = \tfrac{1}{2}(y - x) = -h \qquad \text{(negated)}$$



In the (m, h) coordinates the swap is diagonal:

$$T\,\tau\,T^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

This is the whole secret

Average and difference are the **eigencoordinates of the swap**: $(1, 1)$ has eigenvalue $+1$, $(1, -1)$ has eigenvalue -1. Every later version of the method is this fact, applied to a different swap.

Products and squares in the new coordinates

Substitute $x = m + h$, $y = m - h$:

$$x\,y = (m+h)(m-h) = m^2 - mh + hm - h^2 = m^2 - h^2$$

$$x^2 + y^2 = (m+h)^2 + (m-h)^2 = (m^2 + 2mh + h^2) + (m^2 - 2mh + h^2) = 2m^2 + 2h^2$$

$$x^2 - y^2 = (m+h)^2 - (m-h)^2 = 4\,mh$$

Expressions symmetric in x, y (unchanged by the swap): xy, $x^2 + y^2$, $x + y$, $\max(x,y) + \min(x,y)$, ...

In (m, h) they contain h only through h^2 (or not at all). The terms $\pm mh$, $\pm 2mh$ *cancel*.

Expressions antisymmetric in x, y (negated by the swap): $x - y$, $x^2 - y^2$, $x^3 - y^3$, ...

In (m, h) they are h times something symmetric: $x - y = 2h$, $x^2 - y^2 = 4mh$, $x^3 - y^3 = 2h(3m^2 + h^2)$.

The cancellation, explained

A symmetric expression $f(x,y) = f(y,x)$ becomes $g(m,h) = g(m,-h)$: an *even* function of h, with no odd powers. So every term in $h^1, h^3, \ldots$ must cancel, and it does. The cancellation is not luck; it is what "symmetric" means, seen in the eigencoordinates.

Why the two-unknown problem decouples

The Babylonian problem: find x, y given $x + y = s$ and $xy = P$.

In (x, y): two equations, each involving both unknowns.

$$x + y = s, \qquad xy = P.$$

Substituting gives $x(s - x) = P$; the two unknowns are tangled.

In (m, h): the equations separate.

$$2m = s \implies m = \tfrac{s}{2},$$

$$m^2 - h^2 = P \implies h^2 = \tfrac{s^2}{4} - P.$$

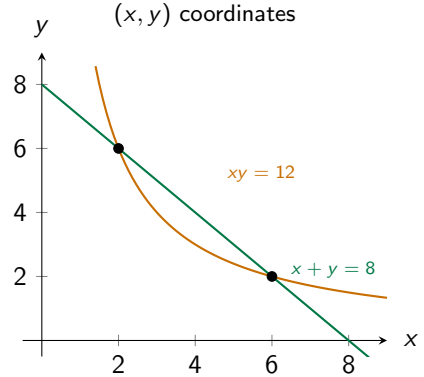
The first involves only m; the second, once m is known, only h^2. Each is one step.

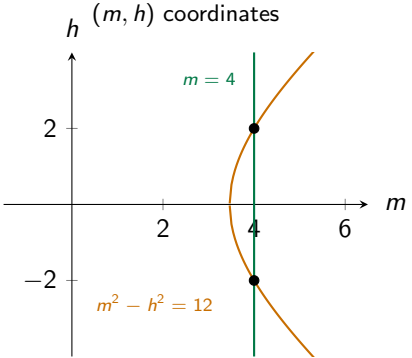
What happened

Both constraints are *symmetric* in x and y, so in the eigencoordinates each is even in h, and the linear one cannot contain h at all. A symmetric linear constraint **fixes** m; a symmetric quadratic constraint then **fixes** h^2. The swap-symmetry of the problem splits it into a "what is shared" part and a "how far apart" part.

The sign of h is not determined, and cannot be: the problem does not distinguish x from y. The two solutions $h = \pm\sqrt{\cdot}$ are the two orderings. The $\pm$ in the quadratic formula is the swap.

The same decoupling, drawn





Left: the line $x + y = 8$ meets the hyperbola $xy = 12$ at $(6, 2)$ and $(2, 6)$ – two points exchanged by the swap. Right: the same two curves after the coordinate change. The line has become *vertical* ($m = 4$: it says nothing about h) and the hyperbola is in standard position, symmetric under $h \mapsto -h$. The intersection is read off as $h = \pm 2$.

Orthogonality and the splitting of "energy"

The vectors $\mathbf{u} = (1, 1)$ and $\mathbf{w} = (1, -1)$ are orthogonal: $\mathbf{u} \cdot \mathbf{w} = 1 - 1 = 0$. Write the point as a combination of them:

$$(x, y) = m\,(1, 1) + h\,(1, -1).$$

Then, by Pythagoras (no cross term, because $\mathbf{u} \perp \mathbf{w}$):

$$x^2 + y^2 = \|(x, y)\|^2 = m^2\|\mathbf{u}\|^2 + h^2\|\mathbf{w}\|^2 = 2m^2 + 2h^2.$$

- The identity from two slides ago, now with a reason: **the mean direction and the difference direction are perpendicular**, so squared length splits into a "mean" share and a "difference" share.
- In statistics: for the sample $\{x, y\}$, $\frac{1}{2}(x^2 + y^2) = m^2 + h^2 = (\text{mean})^2 + (\text{variance})$. "Mean square equals mean squared plus variance" is the orthogonality of $(1, 1)$ and $(1, -1)$. Tuesday's $\|A\|_F^2 = \|S\|_F^2 + \|K\|_F^2$ is the same statement for matrices.

Two identities, two reasons

$xy = m^2 - h^2$ comes from *cancellation of cross terms* (symmetry); $x^2 + y^2 = 2m^2 + 2h^2$ from *orthogonality*. Both follow from the eigencoordinates, but they are not the same fact.

Why exactly two parts, and why $\pm$

Two numbers. There is exactly one way to swap them. The swap τ satisfies $\tau^2 = I$, so its eigenvalues are ± 1 – the square roots of 1. Two eigenvalues, two eigenvectors, two parts.

Three numbers. Write $x_i = m + d_i$ with m the mean and $\sum d_i = 0$. The mean is still a one-dimensional "shared" part. But the deviations (d_1, d_2, d_3) live in a *two*-dimensional space, and permutations of three things do not reduce to a single $\pm$: the deviation space carries a genuinely 2D action of S_3.

The special role of $n = 2$

For n numbers, the "mean" direction $(1, 1, \ldots, 1)$ is always one-dimensional and the deviation space is $(n-1)$-dimensional. Only for $n = 2$ is the deviation space *also* one-dimensional, so that "difference" is a single number and the only symmetry is a sign.

This is why the method is at its cleanest for pairs, and why its later generalizations keep the *pair* structure: a function and its mirror image; a matrix and its transpose; *ab* and *ba*. In every case there are exactly two things and one swap.

From Two Numbers to Anything with a Swap

The hidden pair

Here is the step that makes the two-number case universal.

Observation

Whenever we have an object v and an operation σ with $\sigma^2 = \mathrm{id}$, we have a *pair*: the object and its swapped copy,

$$(\, v,\ \sigma v\,).$$

Applying σ to this pair exchanges its two members: $(v, \sigma v) \mapsto (\sigma v, \sigma^2 v) = (\sigma v, v)$.

So the "two numbers" x, y are now $v, \sigma v$, and everything transfers:

$$v_+ = \tfrac{1}{2}(v + \sigma v), \qquad v_- = \tfrac{1}{2}(v - \sigma v), \qquad v = v_+ + v_-, \qquad \sigma v = v_+ - v_-.$$

- $\sigma v_+ = \frac{1}{2}(\sigma v + v) = v_+$: the average is **fixed** by σ.
- $\sigma v_- = \frac{1}{2}(\sigma v - v) = -v_-$: the half-difference is **negated** by σ.
- The reconstruction $v = v_+ + v_-$ is $x = m + h$; the swapped copy $\sigma v = v_+ - v_-$ is $y = m - h$.

The only difference from the numerical case: v need not be a number – a function, a matrix, a product of vectors, anything that can be added and halved.

The same 2×2 matrix, every time

On the plane spanned by v and σv (assumed independent), in the basis $\{v, \sigma v\}$,

$$\sigma = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \tau,$$

the same matrix as the swap of two numbers, with eigenvectors $v + \sigma v$ $(+1)$ and $v - \sigma v$ (-1).

Setting	**The object** v	**The swap** σ	**The pair** $(v, \sigma v)$
two numbers	(x, y)	$(x, y) \mapsto (y, x)$	(x, y) and (y, x)
functions	f	$f(x) \mapsto f(-x)$	$f(x)$ and $f(-x)$
matrices	A	$A \mapsto A^{\mathsf{T}}$	A and A^{T}
products of vectors	ab	$ab \mapsto ba$	ab and ba

Reading the table

In each row the last column is a pair of "numbers" to average and subtract. The method never changes; what changes is what the two members of the pair *are*, and so what the average and half-difference *mean*.

The projectors, and why they are orthogonal

Write $P_\pm = \frac{1}{2}(\mathrm{id} \pm \sigma)$, so $v_\pm = P_\pm v$. From $\sigma^2 = \mathrm{id}$ alone:

$$P_\pm^2 = P_\pm, \qquad P_+P_- = P_-P_+ = 0, \qquad P_+ + P_- = \mathrm{id}.$$

(Checked on Tuesday.) In the two-number picture these project onto the diagonal and the anti-diagonal.

Orthogonality needs one more ingredient. If the space has an inner product $\langle \cdot, \cdot \rangle$ and σ *preserves* it ($\langle \sigma u, \sigma w \rangle = \langle u, w \rangle$), then

$$\langle v_+, v_- \rangle = 0 \quad \text{and} \quad \|v\|^2 = \|v_+\|^2 + \|v_-\|^2.$$

Proof: $\langle v_+, v_- \rangle = \langle \sigma v_+, \sigma v_- \rangle = \langle v_+, -v_- \rangle$, so it equals its own negative.

Where this holds

- Numbers: the swap preserves $x^2 + y^2$. ✓
- Functions: $\int f(-x)^2\,dx = \int f(x)^2\,dx$. ✓
- Matrices: $\|A^{\mathsf{T}}\|_F = \|A\|_F$. ✓
- Products: $|ba| = |ab|$ in the quaternion and Clifford norms. ✓

Two facts, two hypotheses

Existence and uniqueness of the split needs only $\sigma^2 = \mathrm{id}$; *orthogonality* needs σ to be an isometry. Every swap in this deck is an isometry, so we get both.

The parity rule: how the two parts multiply

With two numbers we saw: symmetric $\times$ symmetric is symmetric, antisymmetric $\times$ antisymmetric is symmetric, symmetric $\times$ antisymmetric is antisymmetric. Signs multiply:

$$(+1)(+1) = +1, \qquad (-1)(-1) = +1, \qquad (+1)(-1) = -1.$$

When the rule holds

If σ respects products, $\sigma(uw) = \sigma(u)\,\sigma(w)$, then for u of parity ϵ_u and w of parity ϵ_w,

$$\sigma(uw) = \sigma(u)\sigma(w) = \epsilon_u\epsilon_w\, uw,$$

so uw has parity $\epsilon_u\epsilon_w$. The decomposition is then a $\mathbb{Z}_2$-**grading**: an even part closed under multiplication, an odd part that flips it.

Example: $x^2 \cdot x^3 = x^5$ – even times odd is odd. $\cos x \sin x$ is odd; $\sin^2 x$ is even. For two numbers, $(x+y)(x-y) = x^2 - y^2$: symmetric times antisymmetric is antisymmetric.

Where the parity rule holds, and where it does not

The rule needs $\sigma(uw) = \sigma(u)\sigma(w)$: the swap must *respect the order* of products.

Holds for:

- Functions under $x \mapsto -x$: $(fg)(-x) = f(-x)g(-x)$. Even $\times$ odd $=$ odd, etc.
- Polynomials: parity of degree.
- Clifford algebra under the *grade involution* (next section).

Fails for:

- Matrices under transpose: $(AB)^{\mathsf{T}} = B^{\mathsf{T}}A^{\mathsf{T}}$ – the *order reverses*.
- Products under $ab \mapsto ba$, for the same reason.

These swaps are **anti**-automorphisms. The parity rule survives only for things that commute.

Why this matters for what follows

The swaps that produce Hamilton's and Clifford's decompositions are anti-automorphisms. So the clean "even times odd is odd" bookkeeping is not available directly, and we have to see what replaces it.

What happens when the swap reverses order

Let σ be an anti-automorphism: $\sigma(uw) = \sigma(w)\sigma(u)$. Take S with $\sigma S = S$ and K with $\sigma K = -K$. Then

$$\sigma(SK) = \sigma(K)\sigma(S) = -KS, \qquad \sigma(KS) = \sigma(S)\sigma(K) = -SK.$$

So neither SK nor KS has a definite parity. But their average and half-difference do:

$$\sigma\big(\tfrac{1}{2}(SK+KS)\big) = -\tfrac{1}{2}(KS+SK) \quad \text{(antisymmetric)}, \qquad \sigma\big(\tfrac{1}{2}(SK-KS)\big) = +\tfrac{1}{2}(SK-KS) \quad \text{(symmetric)}.$$

- The **anticommutator** $\{S, K\} = SK + KS$ of a symmetric and an antisymmetric element is antisymmetric.
- The **commutator** $[S, K] = SK - KS$ of a symmetric and an antisymmetric element is symmetric.
- For two symmetric elements, $\{S_1, S_2\}$ is symmetric and $[S_1, S_2]$ is antisymmetric; likewise for two antisymmetric ones. (This is Exercise 2 from Tuesday.)

The point

When the swap reverses order, the method has to be applied *again*, to the pair (SK, KS), to recover definite parities. Average and difference of a product and its reversal: this is exactly the situation Hamilton and Clifford face, with SK and KS replaced by ab and ba.

Interlude: the three facts, collected

Before moving to products, here is everything the method rests on, in one place.

1. **Eigencoordinates.** For any σ with $\sigma^2 = \mathrm{id}$, the average $\frac{1}{2}(v + \sigma v)$ is fixed by σ and the half-difference $\frac{1}{2}(v - \sigma v)$ is negated. They reconstruct v and σv. *(needs only $\sigma^2 = \mathrm{id}$)*
2. **Cancellation.** Anything built symmetrically from v and σv is even in the half-difference: odd powers of it cancel. This is what decouples problems and kills cross terms. *(needs only $\sigma^2 = \mathrm{id}$ and commutativity of the ingredients)*
3. **Orthogonality.** If σ preserves lengths, the two parts are perpendicular and squared lengths add. *(needs σ to be an isometry)*

And one warning

When the ingredients do not commute, fact 2 must be used with care: the swap may reverse the order of products, and then "symmetric times antisymmetric" is not antisymmetric until you average it with its reversal. Products of vectors are exactly such a case – and that is why Hamilton's and Clifford's products have the shape they have.

Hamilton: The Pair ($\mathbf{uv}$, $\mathbf{vu}$)

Hamilton's product of two vectors

Quaternions: $q = a + b\,\mathbf{i} + c\,\mathbf{j} + d\,\mathbf{k}$ with

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -1, \qquad \mathbf{ij} = \mathbf{k} = -\mathbf{ji}, \quad \mathbf{jk} = \mathbf{i} = -\mathbf{kj}, \quad \mathbf{ki} = \mathbf{j} = -\mathbf{ik}.$$

A *vector* is a quaternion with no scalar part, $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$. Multiply two and sort the result by type:

$$\mathbf{u}\,\mathbf{v} = -(u_1v_1 + u_2v_2 + u_3v_3) + (u_2v_3 - u_3v_2)\,\mathbf{i} + (u_3v_1 - u_1v_3)\,\mathbf{j} + (u_1v_2 - u_2v_1)\,\mathbf{k}.$$

On basis vectors:

$$\mathbf{i}\,\mathbf{i} = -1 \qquad \mathbf{j}\,\mathbf{i} = -\mathbf{k}$$
$$\mathbf{i}\,\mathbf{j} = \mathbf{k} \qquad \mathbf{i}\,\mathbf{i} = -1$$

A vector times itself is a negative scalar. Two different basis vectors give a vector, and swapping them flips its sign.

The pair

The two things to average are $\mathbf{uv}$ and $\mathbf{vu}$. The scalar part is the same in both; the vector part is negated. So

$$\tfrac{1}{2}(\mathbf{uv} + \mathbf{vu}) = -\,\mathbf{u}\cdot\mathbf{v}, \qquad \tfrac{1}{2}(\mathbf{uv} - \mathbf{vu}) = \mathbf{u}\times\mathbf{v}.$$

This is $x = m + h$, $y = m - h$ with $x = \mathbf{uv}$, $y = \mathbf{vu}$.

Polarization: why the symmetric part is a scalar

We do not have to compute the product in coordinates to know what its symmetric part is. Use only one fact: **a vector squared is minus its length squared**, $\mathbf{u}^2 = -|\mathbf{u}|^2$.

Expand $(\mathbf{u}+\mathbf{v})^2$ two ways:

$$(\mathbf{u}+\mathbf{v})^2 = -|\mathbf{u}+\mathbf{v}|^2 = -|\mathbf{u}|^2 - 2\,\mathbf{u}\cdot\mathbf{v} - |\mathbf{v}|^2, \qquad \text{and}$$

$$(\mathbf{u}+\mathbf{v})(\mathbf{u}+\mathbf{v}) = \mathbf{u}^2 + \mathbf{uv} + \mathbf{vu} + \mathbf{v}^2.$$

With $\mathbf{u}^2 = -|\mathbf{u}|^2$ and $\mathbf{v}^2 = -|\mathbf{v}|^2$, comparing gives $\boxed{\mathbf{uv} + \mathbf{vu} = -2\,\mathbf{u}\cdot\mathbf{v}}$, a scalar.

Compare with two numbers:

$$xy = \tfrac{1}{2}\left[(x+y)^2 - x^2 - y^2\right].$$

The "product" is recovered from squares of sums. Same identity, same structure.

What polarization delivers: the symmetric part of the product is determined by the *quadratic form* $\mathbf{u} \mapsto \mathbf{u}^2$ alone. Whatever the rest of the multiplication table says, $\frac{1}{2}(\mathbf{uv} + \mathbf{vu})$ is forced to be $-\mathbf{u}\cdot\mathbf{v}$.

Consequence: if $\mathbf{u} \perp \mathbf{v}$ then $\mathbf{uv} + \mathbf{vu} = 0$, i.e. $\mathbf{uv} = -\mathbf{vu}$. **Orthogonal vectors anticommute.** That is why $\mathbf{ij} = -\mathbf{ji}$.

The antisymmetric part: what is left over

The symmetric part was forced. The antisymmetric part $\frac{1}{2}(\mathbf{uv} - \mathbf{vu})$ is whatever remains, and we can read it off the basis:

$$\tfrac{1}{2}(\mathbf{ij} - \mathbf{ji}) = \tfrac{1}{2}(\mathbf{k} - (-\mathbf{k})) = \mathbf{k}, \qquad \tfrac{1}{2}(\mathbf{ii} - \mathbf{ii}) = 0.$$

Extending bilinearly,

$$\tfrac{1}{2}(\mathbf{uv} - \mathbf{vu}) = (u_2 v_3 - u_3 v_2)\,\mathbf{i} + (u_3 v_1 - u_1 v_3)\,\mathbf{j} + (u_1 v_2 - u_2 v_1)\,\mathbf{k} = \mathbf{u} \times \mathbf{v}.$$

- It is a *vector*, perpendicular to both $\mathbf{u}$ and $\mathbf{v}$, of length $|\mathbf{u}||\mathbf{v}| \sin\theta$.
- It vanishes when $\mathbf{u} \parallel \mathbf{v}$ (the "$h = 0$ when $x = y$" of the two-number case).
- It is negated by the swap, as the half-difference must be: $\mathbf{v} \times \mathbf{u} = -\mathbf{u} \times \mathbf{v}$.

So Hamilton's product is

$$\mathbf{uv} = -\,\mathbf{u} \cdot \mathbf{v} + \mathbf{u} \times \mathbf{v} \qquad \text{(scalar part)} + \text{(vector part)},$$

and the split is nothing but "average and half-difference of $\mathbf{uv}$ and $\mathbf{vu}$." Hamilton wrote $S(\mathbf{uv})$ and $V(\mathbf{uv})$ for the two parts.

A worked example

Let $\mathbf{u} = \mathbf{i} + 2\mathbf{j}$ and $\mathbf{v} = 3\mathbf{i} + \mathbf{k}$.

$$\begin{aligned}\mathbf{uv} &= (\mathbf{i} + 2\mathbf{j})(3\mathbf{i} + \mathbf{k}) = 3\mathbf{ii} + \mathbf{ik} + 6\mathbf{ji} + 2\mathbf{jk} \\ &= -3 - \mathbf{j} - 6\mathbf{k} + 2\mathbf{i} = -3 + (2\mathbf{i} - \mathbf{j} - 6\mathbf{k}) \\ \mathbf{vu} &= (3\mathbf{i} + \mathbf{k})(\mathbf{i} + 2\mathbf{j}) = 3\mathbf{ii} + 6\mathbf{ij} + \mathbf{ki} + 2\mathbf{kj} \\ &= -3 + 6\mathbf{k} + \mathbf{j} - 2\mathbf{i} = -3 - (2\mathbf{i} - \mathbf{j} - 6\mathbf{k})\end{aligned}$$

Average and half-difference:

$$\tfrac{1}{2}(\mathbf{uv} + \mathbf{vu}) = -3,$$

$$\tfrac{1}{2}(\mathbf{uv} - \mathbf{vu}) = 2\mathbf{i} - \mathbf{j} - 6\mathbf{k}.$$

Check against coordinates:

$\mathbf{u}\cdot\mathbf{v} = 1\cdot 3 + 2\cdot 0 + 0\cdot 1 = 3$, $\quad \mathbf{u}\times\mathbf{v} = (2, -1, -6)$.

Pythagoras:

$(\mathbf{u}\cdot\mathbf{v})^2 + |\mathbf{u}\times\mathbf{v}|^2 = 9 + 41 = 50 = 5\cdot 10 = |\mathbf{u}|^2|\mathbf{v}|^2$.

Scalar part and vector part are orthogonal in the quaternion norm, and their squared lengths add to $|\mathbf{uv}|^2 = |\mathbf{u}|^2|\mathbf{v}|^2$: Hamilton's multiplicativity of the norm, i.e. fact 3 for this swap.

The Babylonian identity, in quaternions

Put $m = -\mathbf{u}\cdot\mathbf{v}$ (scalar) and $\mathbf{h} = \mathbf{u}\times\mathbf{v}$ (vector), so $\mathbf{uv} = m + \mathbf{h}$ and $\mathbf{vu} = m - \mathbf{h}$. Multiply them:

$$(\mathbf{uv})(\mathbf{vu}) = \mathbf{u}\,(\mathbf{vv})\,\mathbf{u} = \mathbf{u}\,(-|\mathbf{v}|^2)\,\mathbf{u} = |\mathbf{u}|^2|\mathbf{v}|^2.$$

On the other hand, since m is a scalar and commutes with everything,

$$(m+\mathbf{h})(m-\mathbf{h}) = m^2 - \mathbf{h}^2 = m^2 + |\mathbf{h}|^2 \qquad \text{(because } \mathbf{h}^2 = -|\mathbf{h}|^2 \text{ for a vector)}.$$

So

$$\underbrace{(\mathbf{u}\cdot\mathbf{v})^2}_{m^2} + \underbrace{|\mathbf{u}\times\mathbf{v}|^2}_{-\mathbf{h}^2} = \underbrace{|\mathbf{u}|^2|\mathbf{v}|^2}_{\text{“the product”}}.$$

This is “square of the half-sum minus square of the half-difference equals the product,” $m^2 - h^2 = xy$, with $x = \mathbf{uv}$, $y = \mathbf{vu}$, and $xy = |\mathbf{u}|^2|\mathbf{v}|^2$. The minus sign on h^2 has become a plus because a vector’s square is negative. In trigonometric form: $\cos^2\theta + \sin^2\theta = 1$.

Same identity as YBC 6967, with the two numbers replaced by a product and its reversal.

Which involution is the swap? Quaternion conjugation

The swap $\mathbf{uv} \mapsto \mathbf{vu}$ looks like an operation on *pairs*. It is a single operation on *quaternions*: **conjugation**, $\bar{q} = a - b\,\mathbf{i} - c\,\mathbf{j} - d\,\mathbf{k}$.

- Conjugation reverses products: $\overline{pq} = \bar{q}\,\bar{p}$ (an anti-automorphism), and $\bar{\bar{q}} = q$ (an involution).
- For a vector, $\bar{\mathbf{u}} = -\mathbf{u}$. So for two vectors,

$$\overline{\mathbf{uv}} = \bar{\mathbf{v}}\,\bar{\mathbf{u}} = (-\mathbf{v})(-\mathbf{u}) = \mathbf{vu}.$$

Reversing the order of two vectors is the same as conjugating their product.

Therefore the average/difference of $(\mathbf{uv}, \mathbf{vu})$ is the average/difference of $(q, \bar{q})$ for $q = \mathbf{uv}$:

$\frac{1}{2}(q + \bar{q}) =$ scalar part, $\quad \frac{1}{2}(q - \bar{q}) =$ vector part.

Exactly like $\frac{1}{2}(z + \bar{z}) = \operatorname{Re} z$, $\frac{1}{2}(z - \bar{z}) = i \operatorname{Im} z$ for complex numbers.

What this buys

Hamilton's "scalar and vector of a product" is now a special case of a decomposition defined on *all* quaternions, by a single involution that is an isometry ($|\bar{q}| = |q|$). Facts 1 and 3 apply directly. Fact 2 applies in the commutator/anticommutator form, because conjugation reverses order.

Clifford: The Pair $(ab,\ ba)$

Change one sign

Clifford's modification of Hamilton: let a vector square to *plus* its length squared.

$$a^2 = +|a|^2.$$

With an orthonormal basis $e_1, \ldots, e_n$ of $\mathbb{R}^n$ this gives the rules

$$e_i^2 = 1, \qquad e_i e_j = -e_j e_i \quad (i \neq j),$$

the second of which is *not* an extra assumption: it follows from polarization, exactly as before.

Polarization, third time

$$(a+b)^2 = |a+b|^2 = |a|^2 + 2\,a \cdot b + |b|^2,$$

$$(a+b)(a+b) = a^2 + ab + ba + b^2 = |a|^2 + (ab+ba) + |b|^2,$$

so $\boxed{\frac{1}{2}(ab+ba) = a \cdot b}$. The symmetric part of the product is the dot product, with no minus sign this time. Orthogonal vectors anticommute: $a \cdot b = 0 \Rightarrow ab = -ba$.

Three polarizations

The same computation, three times, in three algebras. In each, the "square of a sum" identity recovers the symmetric part of the product from squares alone.

	two numbers	Hamilton	Clifford
square of a sum	$(x+y)^2 = x^2 + 2xy + y^2$	$(\mathbf{u}+\mathbf{v})^2 = -\lvert\mathbf{u}+\mathbf{v}\rvert^2$	$(a+b)^2 = \lvert a+b\rvert^2$
symmetric part of the product	xy	$-\mathbf{u}\cdot\mathbf{v}$	$a\cdot b$
square of a vector	–	$\mathbf{u}^2 = -\lvert\mathbf{u}\rvert^2$	$a^2 = +\lvert a\rvert^2$
orthogonal vectors	–	anticommute	anticommute

What polarization does not tell you

It fixes the symmetric part only. The antisymmetric part – the half-difference of ab and ba – is whatever the multiplication table says it is, and in each algebra it is a *different kind of object*: a number, a vector, a bivector.

The antisymmetric part: the bivector

Define the **outer product** as the half-difference:

$$\boxed{a \wedge b = \tfrac{1}{2}(ab - ba)}.$$

On the basis of $\mathbb{R}^2$: $\frac{1}{2}(e_1e_2 - e_2e_1) = \frac{1}{2}(e_1e_2 + e_1e_2) = e_1e_2$, and $\frac{1}{2}(e_1e_1 - e_1e_1) = 0$. Bilinearity gives, for $a = a_1e_1 + a_2e_2$ and $b = b_1e_1 + b_2e_2$,

$$a \wedge b = (a_1b_2 - a_2b_1)\, e_1e_2.$$

So Clifford's product is

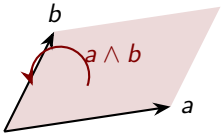
$$ab = a \cdot b + a \wedge b, \qquad a \cdot b = \tfrac{1}{2}(ab + ba), \qquad a \wedge b = \tfrac{1}{2}(ab - ba).$$

Average and half-difference of the pair (ab, ba): the same sentence as for Hamilton, with the sign of the scalar part changed and the vector part replaced by whatever e_1e_2 turns out to be.

What kind of thing is e_1e_2?

$$(e_1e_2)^2 = e_1e_2e_1e_2 = -e_1e_1e_2e_2 = -1.$$

- Not a scalar: it squares to -1.
- Not a vector: it is a product of two vectors and does not lie in $\mathbb{R}^2$.
- A new type, a **bivector**: the oriented plane spanned by e_1 and e_2. Its coefficient in $a \wedge b$, $a_1b_2 - a_2b_1$, is the signed area of the parallelogram on a and b.
- It behaves like i: in $\mathrm{Cl}(2)$, $\{\alpha + \beta\, e_1e_2\}$ is a copy of the complex numbers, and like Hamilton's **k**: $(\mathbf{k})^2 = -1$.



Swapping a and b reverses the circulation: $b \wedge a = -a \wedge b$. The half-difference records *which way around* the two vectors go. That is the information the dot product cannot hold.

A worked example in $\mathbb{R}^2$

Let $a = 2e_1 + e_2$ and $b = e_1 + 3e_2$.

$$ab = (2e_1 + e_2)(e_1 + 3e_2) = 2e_1e_1 + 6e_1e_2 + e_2e_1 + 3e_2e_2 = 5 + 5\,e_1e_2$$
$$ba = (e_1 + 3e_2)(2e_1 + e_2) = 2e_1e_1 + e_1e_2 + 6e_2e_1 + 3e_2e_2 = 5 - 5\,e_1e_2$$

Average and half-difference:

$$a \cdot b = 5, \qquad a \wedge b = 5\,e_1e_2.$$

Check: $a \cdot b = 2 \cdot 1 + 1 \cdot 3 = 5$; signed area $a_1b_2 - a_2b_1 = 6 - 1 = 5$. ✓

The Babylonian identity:

$$(ab)(ba) = a\,(bb)\,a = |b|^2|a|^2 = 10 \cdot 5 = 50,$$

$(m+h)(m-h) = m^2 - h^2 = 25 - (5e_1e_2)^2 = 50$.✓

Here $m = 5$, $h = 5e_1e_2$, and $h^2 = -25$.

In general

$$(a \cdot b)^2 + |a \wedge b|^2 = |a|^2|b|^2 \qquad (\cos^2\theta + \sin^2\theta = 1).$$

Half-sum squared minus half-difference squared (which is negative) equals the product of the pair.

Which involution is the swap? Reversion

As with Hamilton, the swap $ab \mapsto ba$ is a single operation on the whole algebra: **reversion**, $\widetilde{\cdot}$, which reverses the order of every product of vectors: $\widetilde{a_1 a_2 \cdots a_k} = a_k \cdots a_2 a_1$.

- $\widetilde{\widetilde{M}} = M$ (involution); $\widetilde{MN} = \widetilde{N}\,\widetilde{M}$ (anti-automorphism); $|\widetilde{M}| = |M|$ (isometry).
- On scalars and vectors reversion does nothing. On a bivector $e_1 e_2$ it gives $e_2 e_1 = -e_1 e_2$. On a grade-k element it gives the sign $(-1)^{k(k-1)/2}$: $+, +, -, -, +, +, \ldots$

$$\widetilde{ab} = ba \qquad \Longrightarrow \qquad \tfrac{1}{2}(ab + \widetilde{ab}) = a \cdot b \ \text{(grade 0)}, \qquad \tfrac{1}{2}(ab - \widetilde{ab}) = a \wedge b \ \text{(grade 2)}.$$

Two involutions, two jobs

Clifford's algebra carries *both* kinds of involution from the general theory:

- **Reversion** (anti-automorphism): separates $a \cdot b$ from $a \wedge b$. Fact 2 applies in commutator form: for vectors, $\frac{1}{2}(ab - ba)$ is *the* commutator of a and b.
- **Grade involution** α (automorphism, $\alpha(a) = -a$ on vectors): separates even grades from odd. Fact 2 applies in full: even $\times$ even $=$ even, so the even elements form a subalgebra. The product $ab = a \cdot b + a \wedge b$ is entirely even.

Hamilton and Clifford side by side

	Hamilton (1843)	**Clifford (1878)**
square of a vector	$\mathbf{u}^2 = -\lvert\mathbf{u}\rvert^2$	$a^2 = +\lvert a\rvert^2$
the pair	$(\mathbf{uv},\ \mathbf{vu})$	$(ab,\ ba)$
the swap, as an involution	conjugation $q \mapsto \bar{q}$	reversion $M \mapsto \widetilde{M}$
symmetric part (average)	$-\,\mathbf{u}\cdot\mathbf{v}$, a scalar	$a \cdot b$, a scalar
antisymmetric part (half-difference)	$\mathbf{u}\times\mathbf{v}$, a vector	$a \wedge b$, a bivector
square of the antisymmetric part	$-\lvert\mathbf{u}\times\mathbf{v}\rvert^2$	$-\lvert a\wedge b\rvert^2$
Babylonian identity	$(\mathbf{u}\cdot\mathbf{v})^2 + \lvert\mathbf{u}\times\mathbf{v}\rvert^2 = \lvert\mathbf{u}\rvert^2\lvert\mathbf{v}\rvert^2$	$(a\cdot b)^2 + \lvert a\wedge b\rvert^2 = \lvert a\rvert^2\lvert b\rvert^2$
dimensions where it works	3 only	any n

- In $\mathbb{R}^3$ the two are one algebra seen differently: the quaternions are the *even subalgebra* of $\mathrm{Cl}(3)$, with $\mathbf{i}, \mathbf{j}, \mathbf{k}$ the basis bivectors e_2e_3, e_3e_1, e_1e_2. Hamilton's $\mathbf{u}\times\mathbf{v}$ is Clifford's $a\wedge b$ with each plane relabeled by its normal.
- The sign of a^2 is the only real choice. Clifford's makes the metric part $+\,a\cdot b$ and lets the construction run in every dimension, since a bivector needs no normal vector to exist.

Why the decomposition can serve as a definition

Hestenes's presentation (1966), which we followed on Tuesday, *starts* from the geometric product and *defines*

$$a \cdot b := \tfrac{1}{2}(ab + ba), \qquad a \wedge b := \tfrac{1}{2}(ab - ba).$$

Here is why that is legitimate, in terms of the three facts.

1. **Eigencoordinates.** Reversion is an involution, so ab splits uniquely into a reversion-even and a reversion-odd part, and they reconstruct ab and ba. Nothing about the split depends on knowing what the parts *are* in advance.
2. **Cancellation / polarization.** The axiom $a^2 = |a|^2$ determines the symmetric part: it *must* be $a \cdot b$. So the "definition" of the dot product agrees with the one you already had.
3. **Orthogonality.** Reversion is an isometry, so $|ab|^2 = |a \cdot b|^2 + |a \wedge b|^2 = |a|^2|b|^2$: the two parts carry independent information, scalar and oriented-plane, and together they carry all of it.

The one-sentence version

Give me a product of vectors in which $a^2 = |a|^2$; then average and difference of ab and ba *are* the dot and wedge products, and the two-number mechanism guarantees they are well defined, complementary, and orthogonal.

Checking everything in a few lines

The whole chain, numerically, for the $\mathbb{R}^2$ example. A multivector in $\mathrm{Cl}(2)$ is (s, v_1, v_2, b) meaning $s + v_1 e_1 + v_2 e_2 + b\, e_1 e_2$.

```
import numpy as np

def gp(p, q):                          # geometric product in Cl(2)
    s1,x1,y1,b1 = p; s2,x2,y2,b2 = q
    return np.array([ s1*s2 + x1*x2 + y1*y2 - b1*b2,
                      s1*x2 + x1*s2 - y1*b2 + b1*y2,
                      s1*y2 + y1*s2 + x1*b2 - b1*x2,
                      s1*b2 + b1*s2 + x1*y2 - y1*x2 ])

vec = lambda x, y: np.array([0., x, y, 0.])
a, b = vec(2, 1), vec(1, 3)
ab, ba = gp(a, b), gp(b, a)
print(ab, ba)                          # [5 0 0 5]  [5 0 0 -5]
print(0.5*(ab + ba))                   # [5 0 0 0]   a.b  (average)
print(0.5*(ab - ba))                   # [0 0 0 5]   a^b  (half-difference)
print(gp(ab, ba))                      # [50 0 0 0]  = |a|^2 |b|^2  (the "product")
m, h = 0.5*(ab + ba), 0.5*(ab - ba)
print(gp(m, m) - gp(h, h))             # [50 0 0 0]  = m^2 - h^2    (YBC 6967)
```

Summary

One mechanism, four costumes

	Two numbers	**Any involution**	**Hamilton**	**Clifford**
the pair	$x,\ y$	$v,\ \sigma v$	$\mathbf{uv},\ \mathbf{vu}$	$ab,\ ba$
the swap	$(x, y) \mapsto (y, x)$	σ	conjugation	reversion
average m	$\frac{1}{2}(x + y)$	$\frac{1}{2}(v + \sigma v)$	$-\,\mathbf{u} \cdot \mathbf{v}$	$a \cdot b$
half-difference h	$\frac{1}{2}(x - y)$	$\frac{1}{2}(v - \sigma v)$	$\mathbf{u} \times \mathbf{v}$	$a \wedge b$
reconstruction	$x, y = m \pm h$	$v, \sigma v = v_+ \pm v_-$	$\mathbf{uv}, \mathbf{vu} = m \pm \mathbf{h}$	$ab, ba = m \pm h$
what fixes m	a symmetric datum	–	$\mathbf{u}^2 = -\lvert\mathbf{u}\rvert^2$	$a^2 = \lvert a\rvert^2$
"the product" xy	P	–	$(\mathbf{uv})(\mathbf{vu}) = \lvert\mathbf{u}\rvert^2\lvert\mathbf{v}\rvert^2$	$(ab)(ba) = \lvert a\rvert^2\lvert b\rvert^2$
identity $m^2 - h^2 = xy$	$m^2 - h^2 = P$	–	$(\mathbf{u} \cdot \mathbf{v})^2 + \lvert\mathbf{u} \times \mathbf{v}\rvert^2 = \lvert\mathbf{u}\rvert^2\lvert\mathbf{v}\rvert^2$	$(a \cdot b)^2 + \lvert a \wedge b\rvert^2 = \lvert a\rvert^2\lvert b\rvert^2$
orthogonality	$x^2 + y^2 = 2m^2 + 2h^2$	$\lVert v\rVert^2 = \lVert v_+\rVert^2 + \lVert v_-\rVert^2$	$\lvert\mathbf{uv}\rvert^2 = m^2 + \lvert\mathbf{h}\rvert^2$	$\lvert ab\rvert^2 = m^2 + \lvert h\rvert^2$

Reading across a row

Each row is one fact about two numbers and the same fact about products of vectors. The sign of h^2 flips in the last two columns because a vector or bivector squares to a negative scalar; that is the only adjustment.

What to take away

1. **Average and difference are coordinates, not a trick.** They are the (x, y) axes rotated by $45°$. The swap of the two numbers is a reflection, and these are the axes it fixes and flips.
2. **Cancellation is what "symmetric" looks like in those coordinates.** A symmetric expression is even in h, so odd powers of h vanish. This is why a symmetric linear constraint fixes m, a symmetric quadratic constraint fixes h^2, and the Babylonian problem decouples.
3. **Any involution supplies a pair.** The thing and its swapped copy play the roles of x and y. Functions, matrices, products, states – the same 2×2 matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ every time.
4. **Order-reversing swaps need the method applied to products.** Transpose, conjugation, and reversion reverse the order of factors. The average/difference of ab and ba is how you recover definite parities, and that is *why* Hamilton's and Clifford's products split into a scalar and a non-scalar.
5. **Polarization pins down the symmetric part.** Once a vector's square is its length squared (up to sign), $\frac{1}{2}(ab + ba)$ *must* be the dot product. The wedge is what remains. Nothing else is needed to define either.

Exercises

1. Two numbers have sum 14 and sum of squares 106. Find them using (m, h) coordinates *without* solving a quadratic in x. Which constraint fixed m, which fixed h^2?
2. Show directly that $x^3 + y^3$ is a polynomial in m and h^2, and that $x^3 - y^3$ is h times a polynomial in m and h^2. Then state the general rule and prove it from $f(x, y) = f(y, x)$.
3. For three numbers x_1, x_2, x_3 with mean m and deviations d_i: show $\sum x_i^2 = 3m^2 + \sum d_i^2$. Why is there no cross term? Which part of the two-number argument does this use, and which part does it *not* generalize?
4. Let σ be an involution that is *not* an isometry (e.g. $\sigma(x, y) = (2y, x/2)$ on $\mathbb{R}^2$). Verify $\sigma^2 = \mathrm{id}$, find v_+ and v_- for $v = (1, 0)$, and check whether $\|v\|^2 = \|v_+\|^2 + \|v_-\|^2$. Which of the three facts fails?

Exercises, continued

5. In the quaternions, compute $\mathbf{uv}$ and $\mathbf{vu}$ for $\mathbf{u} = \mathbf{i} + \mathbf{j}$, $\mathbf{v} = \mathbf{j} + \mathbf{k}$. Verify $(\mathbf{u} \cdot \mathbf{v})^2 + |\mathbf{u} \times \mathbf{v}|^2 = |\mathbf{u}|^2 |\mathbf{v}|^2$. Then verify $\overline{\mathbf{uv}} = \mathbf{vu}$.
6. In $\mathrm{Cl}(3)$ with $a = e_1 + e_2$ and $b = e_2 + e_3$, compute ab, ba, $a \cdot b$, $a \wedge b$, and $(ab)(ba)$. Then compute $(a \wedge b)^2$ and confirm $(a \cdot b)^2 - (a \wedge b)^2 = |a|^2 |b|^2$.
7. Let S be a symmetric matrix and K antisymmetric. Using only $(AB)^{\mathsf{T}} = B^{\mathsf{T}} A^{\mathsf{T}}$, determine the parity of $SK + KS$, $SK - KS$, K^2, and SKS. Which of these would come out differently if transpose were an automorphism instead of an anti-automorphism?
8. (Harder.) Reversion and the grade involution are both involutions on $\mathrm{Cl}(n)$. Their composite, Clifford conjugation $\bar{M} = \widetilde{\alpha(M)}$, is a third. Tabulate the signs of all three on grades 0–4, and identify which of the three gives the decomposition $ab \mapsto (a \cdot b, a \wedge b)$ and which gives $q \mapsto (\text{scalar}, \text{vector})$ when $\mathrm{Cl}^+(3)$ is identified with the quaternions.

Next

- With the mechanism in hand, the FGA unit can treat $a \cdot b$ and $a \wedge b$ as *derived* from one product, and ask the semantic question directly: for two word vectors, what does the average of ab and ba represent, and what does their half-difference represent?
- The answer offered on Tuesday – likeness and oriented relation – is now backed by the three facts: the two parts are unique, complementary, and orthogonal, and the second one is the only place the *order* of the two words can be recorded.
- The attention probing question (how much of $W_Q^{\mathsf{T}} W_K$ is in K) is the matrix row of the big table; the FGA role/filler question is the Clifford row. Same mechanism, read in two settings.

Reading

- Any linear algebra text: eigenvectors of a reflection; projections onto complementary subspaces.
- Hestenes and Sobczyk, *Clifford Algebra to Geometric Calculus*, §1.1–1.2 (the axioms and the inner/outer split).
- Dorst, Fontijne & Mann, *Geometric Algebra for Computer Science*, §6.1–6.3.
- Course notes, *Geometric Algebra as a Programming Language*, Unit 2 (“Measurement”).